

WORKFROMHOME

Digital Forensics & Incident Response Investigation

Windows Forensic Investigation Report

Mehdi Barbereau

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Based on the CyberDefenders WorkFromHome training lab

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AI Assistance Disclosure

The investigation, artifact analysis, and incident reconstruction were conducted by the author. AI was used to draft and refine the report based on the author's findings. The author reviewed the content and remains responsible for its accuracy and conclusions.

Project Overview

This project presents a Digital Forensics and Incident Response (DFIR) investigation of a compromised Windows environment based on the CyberDefenders WorkFromHome lab.

The objective of the investigation was to reconstruct the incident timeline, determine how the threat actor gained and expanded access to the environment, identify compromised data and systems, and analyze the techniques used throughout the intrusion.

The investigation involved the analysis and correlation of multiple Windows forensic artifacts, including Windows Event Logs, NTFS metadata, Prefetch files, Registry data, and browser artifacts. These artifacts were used to establish a chronological sequence of events and support the investigation findings with forensic evidence.

The investigation was conducted using several forensic tools, primarily the Eric Zimmerman toolset, including MFTECmd, PECmd, EvtxECmd, Timeline Explorer, and Registry Explorer. Event Log Explorer and Windows Event Viewer were used for Windows event log analysis, while DB Browser for SQLite was used to examine SQLite-based artifacts such as browser data. Notepad++ was also used for reviewing and inspecting text-based evidence.

Rather than focusing solely on individual indicators, the analysis aims to reconstruct the complete attack chain, correlate evidence across multiple forensic sources, and provide evidence-based conclusions for each stage of the incident.

Lab: CyberDefenders — WorkFromHome

Difficulty: Medium

Category: Digital Forensics & Incident Response (DFIR)

Environment: Windows

Tools: Eric Zimmerman Tools · Event Log Explorer · Event Viewer · DB Browser for SQLite · Registry Explorer · Notepad++

Executive Summary

The incident began when a junior developer downloaded the remote access application AnyDesk and granted remote access to an external actor using the profile name ZeroCTO.

After gaining access to the workstation, the attacker discovered and accessed internal resources. The attacker subsequently obtained access to a sensitive file named `Annual_Business_Plan_2025.pdf`, downloaded it to the compromised system, and transferred the data to their own system.

The attacker later compromised a senior developer's account and successfully accessed it via RDP. After gaining access, the attacker abused elevated privileges to modify the Windows `sethc.exe` executable and obtain a privileged shell.

Finally, the attacker added the AnyDesk directory to Microsoft Defender exclusions to reduce security detection and configured AnyDesk to start automatically at system startup, establishing persistent remote access to the compromised system.

Incident Timeline

Time	Event	Evidence Source
09:02	AnyDesk.exe was downloaded to the compromised workstation	Download artifact
09:03	A remote user operating under the profile name ZeroCTO was granted access to the workstation through AnyDesk.	AnyDesk / communication artifact
09:10	The attacker accessed the internal website http://file.zerodivine-u.com/ using previously compromised senior developer credentials.	Microsoft Edge History — urls table
09:10	Annual_Business_Plan_2025.pdf was downloaded from the internal resource to the compromised workstation.	Browser / download artifact
09:12	The collected business document was transferred from the compromised workstation to the threat actor's system.	File transfer artifact
09:14	The attacker attempted to authenticate to the senior developer's account from 192.168.189.129.	Authentication artifact
09:15	Authentication succeeded and an RDP session was established using the senior developer's account.	RDP / authentication artifact
09:15	The compromised account was identified as possessing SeTakeOwnershipPrivilege.	Privilege artifact
09:18	takeown.exe and icacls.exe were used against C:\Windows\System32\sethc.exe.	Filesystem / execution artifacts
09:20	The legitimate sethc.exe was renamed to sethc.exe.bak.	Filesystem artifact
09:20	cmd.exe was copied to C:\Windows\System32\sethc.exe, replacing the Sticky Keys executable.	Filesystem artifact
09:22	The modified sethc.exe was executed and a new elevated session was obtained.	Execution / session artifact
09:25	C:\Program Files (x86)\Anydesk was added to Microsoft Defender's antivirus exclusions.	Microsoft Defender configuration artifact
09:31	AnyDesk was configured to execute automatically at system startup, establishing persistence.	Service / persistence artifact

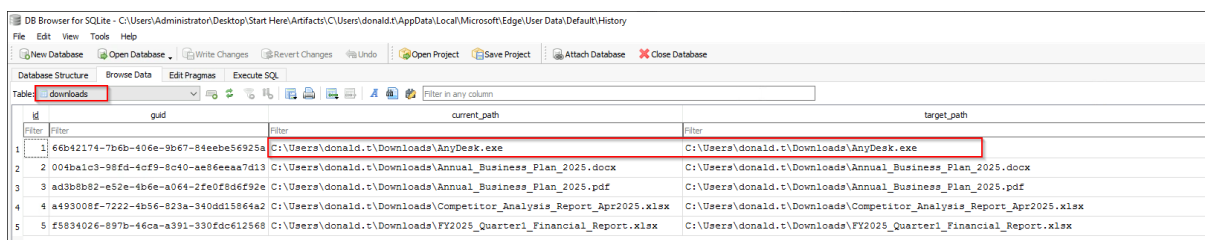
Table 1 — Reconstructed incident timeline based on the analysed forensic artifacts

Technical Investigation

Initial Access — AnyDesk

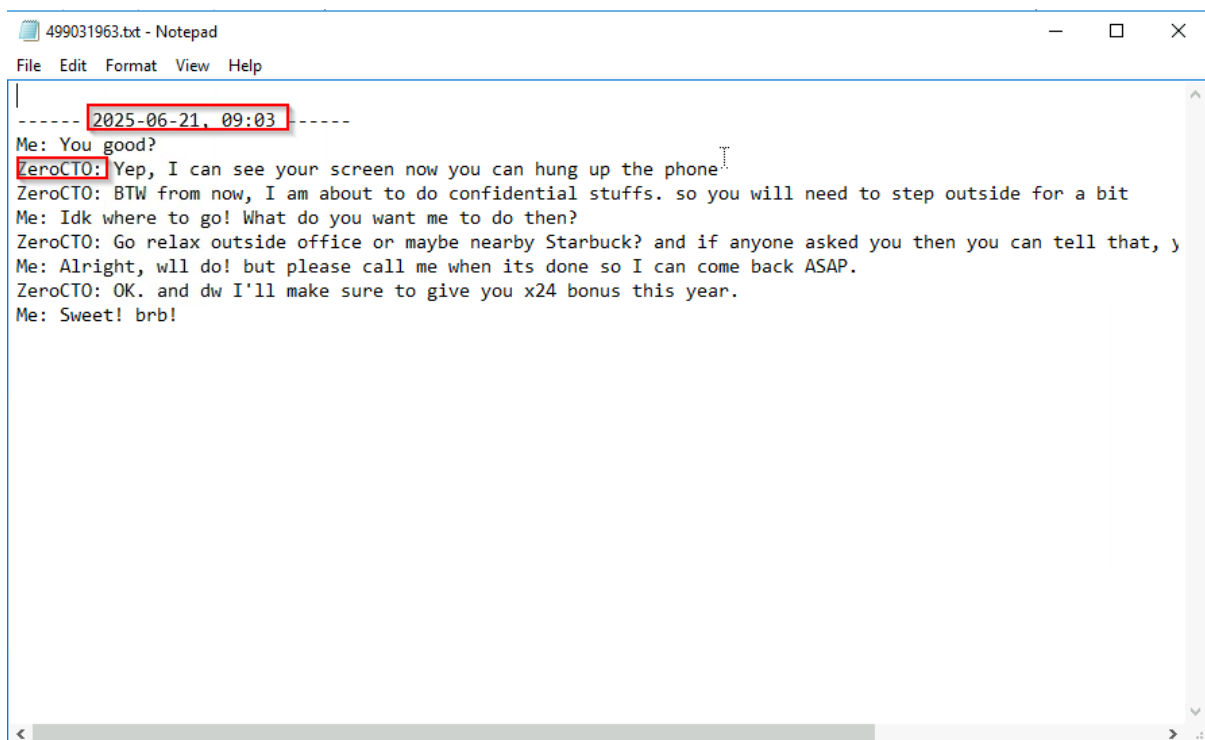
Evidence

Analysis of the available forensic artifacts identified the download of the remote access application AnyDesk on June 21, 2025 at approximately 09:02. The artifact shows that AnyDesk.exe was downloaded to the compromised workstation.



	id	guid	current_path	target_path
1	66b42174-7b6b-406e-9b67-84eebe56925a		C:\Users\donald.t\Downloads\AnyDesk.exe	C:\Users\donald.t\Downloads\AnyDesk.exe
2	004ba1c3-98fd-4cf9-8c50-a86eeaa7d13		C:\Users\donald.t\Downloads\Annual_Business_Plan_2025.docx	C:\Users\donald.t\Downloads\Annual_Business_Plan_2025.docx
3	ad3b8b82-e52e-4b6e-a064-2fe0f8d6f92e		C:\Users\donald.t\Downloads\Annual_Business_Plan_2025.pdf	C:\Users\donald.t\Downloads\Annual_Business_Plan_2025.pdf
4	a493008f-7222-4b56-823a-340dd15864a2		C:\Users\donald.t\Downloads\Competitor_Analysis_Report_Apr2025.xlsx	C:\Users\donald.t\Downloads\Competitor_Analysis_Report_Apr2025.xlsx
5	f5834026-897b-46ca-a391-330fde612568		C:\Users\donald.t\Downloads\FY2025_Quarter1_Financial_Report.xlsx	C:\Users\donald.t\Downloads\FY2025_Quarter1_Financial_Report.xlsx

Further analysis identified an AnyDesk connection approximately one minute later. At 09:03, a remote user operating under the profile name ZeroCTO was granted access to the workstation.



```
499031963.txt - Notepad
File Edit Format View Help

----- 2025-06-21, 09:03 -----
Me: You good?
ZeroCTO: Yep, I can see your screen now you can hung up the phone
ZeroCTO: BTW from now, I am about to do confidential stuffs. so you will need to step outside for a bit
Me: Idk where to go! What do you want me to do then?
ZeroCTO: Go relax outside office or maybe nearby Starbuck? and if anyone asked you then you can tell that, y
Me: Alright, wll do! but please call me when its done so I can come back ASAP.
ZeroCTO: OK. and dw I'll make sure to give you x24 bonus this year.
Me: Sweet! brb!
```

Analysis

The close timing between the download of AnyDesk.exe and the subsequent remote connection indicates that AnyDesk was used as the remote access mechanism during the initial stage of the compromise.

The connection associated with ZeroCTO provided the external actor with interactive access to the workstation, creating an entry point for the activity observed later in the investigation.

The available forensic evidence establishes the download of AnyDesk and the subsequent remote connection. However, the analyzed artifacts do not establish why the user downloaded AnyDesk or how the external actor convinced the user to grant access. Therefore, social engineering cannot be independently confirmed from the available evidence.

Internal Resource Access and Data Collection

Evidence

Analysis of the Microsoft Edge History database, specifically the urls table, identified access to the internal website <http://file.zerodivine-u.com/> at approximately 09:10.

According to the incident context provided by the lab, the threat actor had previously obtained credentials from a senior developer through social engineering. The credential acquisition itself was not reconstructed from the forensic artifacts analyzed in this investigation.

Shortly after accessing the internal resource, the attacker downloaded `Annual_Business_Plan_2025.pdf` to the compromised workstation.

Analysis

The Edge browser history establishes that the internal website was accessed at approximately 09:10, after the attacker had obtained the senior developer's credentials.

This demonstrates that the attacker progressed from remote access to the initial workstation to accessing an internal organizational resource using compromised credentials.

The subsequent download of `Annual_Business_Plan_2025.pdf` indicates that sensitive information from the internal resource was collected on the compromised workstation.

Based on the available forensic evidence and the incident context provided by the lab, the attacker used compromised senior developer credentials to access an internal resource and subsequently collect sensitive organizational data.

Data Exfiltration

Evidence

At approximately 09:12, forensic artifacts indicate that the previously downloaded data was transferred from the compromised workstation to the attacker's system.



Analysis

The transfer occurred shortly after the attacker downloaded Annual_Business_Plan_2025.pdf from the internal resource.

The sequence of events establishes a progression from internal resource access to local collection of the document, followed by its transfer away from the compromised workstation.

This activity represents the direct confidentiality impact of the incident because organizational data was transferred outside the compromised environment.

Based on the available evidence, data collected from the internal resource was exfiltrated from the compromised workstation to the threat actor's system.

Senior Developer Account Compromise and RDP Access

Evidence

At approximately 09:14, the attacker attempted to authenticate to the senior developer's account from the compromised system with the following source IP address:

192.168.189.129

At approximately 09:15, the authentication was successful and the attacker established an RDP session using the senior developer's account.

Successful logon	WORKGROUP\ZEROB\$	ZEROB (192.168.189.129)	Target: ZEROB\john.k
Successful logon	WORKGROUP\ZEROB\$	ZEROB (192.168.189.129)	Target: ZEROB\john.k

Analysis

The successful RDP connection demonstrates that the attacker expanded access beyond the initially compromised user context.

The progression from the observed authentication attempt to the successful RDP session is consistent with the attacker using compromised credentials to obtain interactive access under the senior developer's account.

This represented an important escalation in the incident because subsequent forensic evidence shows that this account possessed privileges that could be abused to manipulate protected Windows resources.

Based on the available evidence, the attacker successfully used the senior developer's account to establish an RDP session and expand access within the compromised environment.

Privilege Escalation — Sticky Keys Backdoor

Evidence

At approximately 09:15, analysis revealed that the compromised account possessed SeTakeOwnershipPrivilege.

This Windows privilege allows an account to take ownership of securable objects, including files, potentially enabling subsequent modification of their permissions.

At approximately 09:18, two legitimate Windows utilities were observed targeting sethc.exe:

takeown.exe

icaccls.exe

The targeted executable was:

C:\Windows\System32\sethc.exe

The legitimate file was protected by TrustedInstaller.

Line	Tag	Update Timestamp	Name	Update Reasons
		2025-06-21 00:00:00	exe	SecurityChange
88173		2025-06-21 09:15:49	python.exe	FileCreate SecurityChange RepairPointChange
88174		2025-06-21 09:15:49	python.exe	FileCreate SecurityChange RepairPointChange Close
88177		2025-06-21 09:15:49	python.exe	FileCreate SecurityChange RepairPointChange
88178		2025-06-21 09:15:49	python.exe	FileCreate SecurityChange RepairPointChange Close
94527		2025-06-21 09:17:19	OneDriveSetup.exe	FileCreate EaChange SecurityChange
94528		2025-06-21 09:17:19	OneDriveSetup.exe	DataExtend FileCreate EaChange SecurityChange
94529		2025-06-21 09:17:19	OneDriveSetup.exe	DataOverwrite DataExtend FileCreate EaChange SecurityChange
94530		2025-06-21 09:17:19	OneDriveSetup.exe	DataOverwrite DataExtend FileCreate EaChange SecurityChange BasicInfo
94531		2025-06-21 09:17:19	OneDriveSetup.exe	DataOverwrite DataExtend FileCreate EaChange SecurityChange BasicInfo
100660		2025-06-21 09:18:41	sethc.exe	SecurityChange
100661		2025-06-21 09:18:41	sethc.exe	SecurityChange Close
100671		2025-06-21 09:18:51	sethc.exe	SecurityChange
100672		2025-06-21 09:18:51	sethc.exe	SecurityChange Close
102692		2025-06-21 09:20:42	sethc.exe	FileCreate EaChange SecurityChange
102693		2025-06-21 09:20:42	sethc.exe	DataExtend FileCreate EaChange SecurityChange
102694		2025-06-21 09:20:42	sethc.exe	DataOverwrite DataExtend FileCreate EaChange SecurityChange
102695		2025-06-21 09:20:42	sethc.exe	DataOverwrite DataExtend FileCreate EaChange SecurityChange BasicInfo
102696		2025-06-21 09:20:42	sethc.exe	DataOverwrite DataExtend FileCreate EaChange SecurityChange BasicInfo
226241		2025-06-21 11:17:37	API_BASE.EXE-3F780C95.pf	SecurityChange
226242		2025-06-21 11:17:37	API_BASE.EXE-3F780C95.pf	SecurityChange Close
226243		2025-06-21 11:17:37	API_DELTA.EXE-78CAB780.pf	SecurityChange
226244		2025-06-21 11:17:37	API_DELTA.EXE-78CAB780.pf	SecurityChange Close
226245		2025-06-21 11:17:37	API_ENGINE.EXE-F1C956E4.pf	SecurityChange
226246		2025-06-21 11:17:37	API_ENGINE.EXE-F1C956E4.pf	SecurityChange Close

Further filesystem analysis established that at approximately **09:20**, the legitimate Sticky Keys executable was renamed:

sethc.exe → sethc.exe.bak

The attacker subsequently copied:

C:\Windows\System32\cmd.exe

to:

C:\Windows\System32\sethc.exe

As a result, the legitimate Sticky Keys executable was replaced by a copy of the Windows Command Prompt.

At approximately 09:22, the modified sethc.exe was executed and a new elevated session was obtained.

Volume	Source Created	Source Modified	Source Accessed	Executable Name	Hash	Size	Version	Last Run	Previous Run	Priority
	2026-09-26 15:56:25	2025-06-21 0...	2026-09-26 15...	sethc	2 1E0000A0	9890	Windows ...	2025-06-21 09:27:45	2025-06-21 09:22:27	

Analysis

The artifacts reconstruct a clear privilege-escalation sequence.

First, the attacker identified that the compromised account possessed SeTakeOwnershipPrivilege. The attacker then used the legitimate Windows utilities takeown.exe and icaccls.exe against the protected sethc.exe executable.

The original executable was preserved as sethc.exe.bak, while cmd.exe was copied into its original location. This caused the Sticky Keys functionality to launch a command shell instead of the legitimate accessibility application.

The subsequent execution of the modified sethc.exe at approximately **09:22**, followed by the elevated session, supports the conclusion that the modification was successfully used to obtain higher privileges.

Based on the available evidence, the attacker abused SeTakeOwnershipPrivilege and legitimate Windows utilities to replace sethc.exe with cmd.exe and obtain a privileged command shell.

Defense Evasion — Microsoft Defender Exclusion

Evidence

At approximately 09:25, Microsoft Defender configuration artifacts showed that the following directory was added to the antivirus exclusion configuration:

C:\Program Files (x86)\Anydesk

The modification occurred after the attacker had obtained elevated access to the system.

Time Created	Event Id	Level	Category	Source	Target	Paylo.	Payload Data2
2025-06-21 09:25:16	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\Exclusions\Paths\C:\Program Files (x86)\Anydesk = 0x0
2025-06-21 11:40:09	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\UIS\Consumers\IPS\DisableBleNetworkSensor = 0x1
2025-06-21 11:40:25	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\IsServiceRunning = 0x1
2025-06-21 11:40:25	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\Diagnostics\InitializingComponentProgress = PostPlatformUpdate
2025-06-21 11:40:26	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\ServiceStartStates = 0x1
2025-06-21 11:40:26	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\Diagnostics\InitializingComponentProgress = LoadingEngine
2025-06-21 11:40:26	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\Features\TPExclusions = 0x0
2025-06-21 11:40:42	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\SAC\ClientWaitForRpc = 0x0
2025-06-21 11:40:42	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\UXPFlags = 0x0
2025-06-21 11:40:42	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\ProductAppDataPath = C:\ProgramData\Microsoft\Windows Defender
2025-06-21 11:40:42	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\CoreService\WdTimerInitialDelay = 0x493E2
2025-06-21 11:40:42	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\CoreService\WdDisableResController = 0x0
2025-06-21 11:40:42	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\CoreService\WdTrustedRootCertThumbPrints = CB3CC8B76031E5E0138F80D39A23
2025-06-21 11:40:42	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\CoreService\WdAlertMonitorWindow = 0x19
2025-06-21 11:40:42	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\CoreService\WdEnableDailySensorChecks = 0x0
2025-06-21 11:40:42	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\CoreService\WdConfigHash = 0x6935666
2025-06-21 11:40:42	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\CoreService\WdTrustedSubjectOrgs = Microsoft Corporation\Digicert Inc
2025-06-21 11:40:42	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\CoreService\WdTimerMonitorInterval = 0x493E0
2025-06-21 11:40:42	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\CoreService\WdAlertMinInterval = 0x3C
2025-06-21 11:40:42	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\CoreService\WdEnableCoreSvcRemediation = 0x0
2025-06-21 11:40:42	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\CoreService\CpuSensor\HighThresholds = 45 95 20
2025-06-21 11:40:42	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\CoreService\CpuSensor\MonitoredTargets = mpdefendercoreservice[mspeng]
2025-06-21 11:40:42	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\CoreService\CpuSensor\LowThresholds = 45 95 20
2025-06-21 11:40:42	5007	Info	Windows Defender	Windows Defender	Windows Defender	Old V.	New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\CoreService\CrashSensor\MonitoredTargets =

Analysis

Adding a directory to Microsoft Defender's exclusion list reduces antivirus inspection of content located within that path.

In the context of the reconstructed attack timeline, the modification is significant because it specifically targeted the directory associated with AnyDesk, the remote access software used earlier in the incident.

The timing is also relevant: the Defender configuration change occurred after the attacker obtained elevated access and shortly before the persistence activity identified later in the investigation.

Based on the available evidence, the attacker modified Microsoft Defender's configuration to exclude the AnyDesk directory from antivirus scanning, providing defense evasion for activity associated with the remote access software.

Persistence — AnyDesk Auto-Start

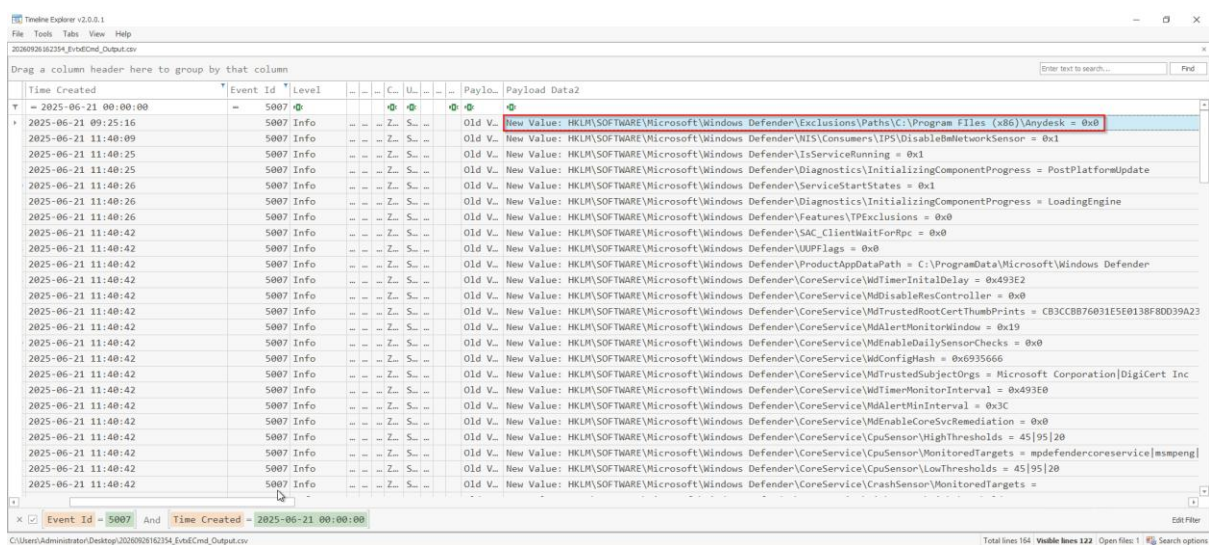
Evidence

At approximately 09:31, the investigation identified a persistent AnyDesk component.

The persistence entry was configured to launch anydesk.exe automatically when the system starts and contained the following description:

Start anydesk on startup

This activity occurred approximately six minutes after the AnyDesk directory had been added to Microsoft Defender exclusions.



anydesk									
Name	Description	Display Name	Start Mode	Service Type	Name Key Last Write	Parameters Key Last ...	Group	Image Path	Service DLL
AnyDesk	Start Anydesk on startup	AnyDesk	Automatic	Win32OwnProcess	2025-06-21 09:31:04			C:\Program Files (x86)\AnyDesk\AnyDesk.exe	

Analysis

Configuring AnyDesk to execute automatically at system startup allowed the remote access software to remain available following a reboot.

This activity is distinct from the Defender exclusion identified in the previous section. The Defender modification provided defense evasion, whereas the automatic execution of AnyDesk provided persistence.

The reconstructed sequence is therefore:

Privileged Access → Defender Exclusion → AnyDesk Auto-Start

Together, these artifacts demonstrate that after obtaining elevated access, the attacker modified the system's security configuration and subsequently established a mechanism for continued remote access.

Based on the available evidence, the attacker established persistence by configuring AnyDesk to start automatically with the system, enabling continued remote access following a reboot.

MITRE ATT&CK Mapping

The attacker behaviors identified during the forensic investigation were mapped to the MITRE ATT&CK Enterprise framework. The mapping below includes techniques supported by the available forensic evidence and incident context.

Tactic	Technique	ID	Investigation Evidence
Initial Access	Valid Accounts	T1078	Compromised senior developer credentials were used to access internal resources.
Command and Control	Remote Desktop Software	T1219.002	AnyDesk provided ZeroCTO with interactive remote access.
Lateral Movement	Remote Desktop Protocol	T1021.001	An RDP session was established using the compromised senior developer account
Privilege Escalation	Accessibility Features	T1546.008	sethc.exe was replaced with cmd.exe to obtain a privileged shell.
Defense Impairment	Disable or Modify Tools	T1685	The AnyDesk directory was added to Microsoft Defender exclusions.
Persistence	Windows Service	T1543.003	AnyDesk was configured to start automatically with Windows.

Indicators of Compromise and Relevant Artifacts

Type	Indicator / Artifact	Relevance
IP Address	192.168.189.129	Source IP associated with authentication activity targeting the senior developer account.
Internal Domain	file.zerodivine-u.com	Internal resource accessed using compromised senior developer credentials.
Remote Access Software	AnyDesk.exe	Remote access software used by ZeroCTO to interact with the compromised workstation.
Defender Exclusion Path	C:\Program Files (x86)\Anydesk	Directory added to Microsoft Defender exclusions after elevated access was obtained.
Modified System File	C:\Windows\System32\sethc.exe	Sticky Keys executable replaced with a copy of cmd.exe during privilege escalation.
Backup Artifact	C:\Windows\System32\sethc.exe.bak	Backup of the legitimate Sticky Keys executable created during the modification.

Recommendations

Priority	Recommendation	Rationale
High	Restrict Unauthorized Remote Access Software	Implement application control policies to prevent unauthorized remote access tools such as AnyDesk from being installed or executed. Approved remote access software should be centrally managed and monitored.
High	Monitor Microsoft Defender Configuration Changes	Generate alerts when antivirus exclusions are created or modified, particularly for user-controlled directories or remote access software. Defender configuration changes should be restricted to authorized administrators.
High	Strengthen RDP Access Controls	Restrict RDP access to systems and accounts that require it, enforce MFA where applicable, and monitor successful and failed remote logons for unusual source systems or accounts.
High	Protect Privileged Accounts	Apply least privilege to administrative and privileged accounts and review privileges such as SeTakeOwnershipPrivilege. Privileged credentials should not be used for routine activity.
Medium	Monitor Critical Windows File Modifications	Detect modifications to protected system and accessibility executables such as sethc.exe, including ownership changes, permission modifications, renaming, or replacement with command interpreters.
Medium	Monitor Persistence Mechanisms	Monitor the creation and modification of Windows services and other auto-start mechanisms, particularly when remote access software is configured to execute automatically at system startup.

Conclusion

This investigation reconstructed the compromise of the Windows environment by correlating multiple forensic artifacts across the incident timeline. The evidence established a progression from unauthorized remote access through AnyDesk to internal resource access, data exfiltration, compromised-account use through RDP, privilege escalation, defense evasion, and persistence.

The investigation also demonstrated how legitimate Windows features and remote access software can be abused during an intrusion. Correlating browser history, Windows artifacts, filesystem activity, security configuration changes, and persistence mechanisms provided the evidence necessary to reconstruct the attack chain and assess its impact.

The identified findings and recommended security controls can be used to improve monitoring and reduce the likelihood of similar activity succeeding in the future.